LEARNING MODULE

30

Estimation Error and Robust Portfolios

An optimizer treats every estimate as exact: measure which errors break the plug-in portfolio, and test the repairs against equal weights.

VOLUME V · Portfolio Optimization

BUILDS ON LM7 · LM9 · LM10 · LM11 · LM12 · LM14 · LM15 · LM16 · LM18 · LM20 · LM22 · LM23 · LM24 · LM26 · LM27 · LM28 · LM29

LEADS TO LM31 · LM32 · LM33 · LM34 · LM35 · LM45

TIME ≈ 3 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	1 simulate plug-in portfolios from a known model and measure the loss due to estimated means
☐	2 explain why equal weights are hard to beat, and state the conditions of the 1/N evidence
☐	3 derive the oracle shrinkage weight as a bias–variance trade-off
☐	4 compute Ledoit–Wolf shrinkage by hand and with scikit-learn, and diagnose its effect on eigenvalues and weights
☐	5 derive the long-only minimum-variance portfolio as an unconstrained one on an adjusted covariance matrix
☐	6 derive the Black–Litterman posterior for one view and compute the weights it implies
☐	7 choose a repair (shrinkage, constraints, a reference portfolio, a factor model or resampling) for a given ratio of observations to assets
☐	8 compare out-of-sample volatilities of sample, shrunk and constrained portfolios, labelling each result

LEARNING MODULE OVERVIEW

- With the first window's estimates as the truth, the plug-in tangency portfolio keeps a mean true Sharpe ratio of 0.423 of the possible 0.880; with the means known it would keep 0.87686.
- Errors in the means alone give the exposure to 00646 a standard deviation of 65.7 points; errors in the covariances alone, 4.73 points.
- A mean's standard error falls only with years: two years of sessions leave 8.2 points for 00646, however many days they hold.
- Shrinkage toward $\bar{\lambda} I$ keeps the eigenvectors and moves the eigenvalues. On 60 returns of 28 sectors it cuts the condition number from 438 to 68.1 and the realized volatility from 13.85% to 12.04%.
- With 504 days for five funds shrinkage hardly matters: out of sample, R5 realized 9.97% against R3's 9.96%.
- The long-only minimum-variance portfolio is the unconstrained one of an adjusted matrix, (30.4). On U28's first half 19 constraints bind, and the second half's volatility rose from 11.78% to 12.14%.
- Black–Litterman starts from R2's implied returns; one view at $c = 0.5$ moves only 0050 and 00646, to 22.5% and 37.5%.
- Resampled weights hold every fund, from 3.8% to 35.2%, although each fund is excluded in 18% to 72% of the resamples.

1 INTRODUCTION

LM28 and LM29 solved for weights as if the means $\boldsymbol{\mu}$ and the covariance matrix Σ were known. In practice both are estimated. This module measures the error in the inputs, finds which error matters, and tests the repairs.

MOTIVATING CASE **Five funds, two years, and positions nobody should hold**

On 27 February 2019 the through case reads its first window: 504 daily returns of five funds. Fed the window's means and covariances, the tangency formula of LM28 puts 62.6% in 00646 and shorts 14.7% of 00642U. Its in-sample Sharpe ratio is 0.88, against 0.33 for equal weights.

Now suppose those estimates were the truth, and draw another 504 days from them, 10,000 times. Re-estimated on the new days, the position in 00646 has a standard deviation of 67.1 points. The portfolio's true Sharpe ratio averages 0.423. If only the covariances had to be estimated, it would average 0.87686, close to the possible 0.880. The simulation's standard errors are 0.5 points, 0.003 and 0.00002.

Out of sample the through case agrees that the covariances are estimated well enough. From March 2019 to October 2026, minimum variance, which uses no means, realized 9.96% volatility net of costs, against 12.48% for equal weights. Shrinking its covariance matrix changed that by 0.01 points. Which errors matter, and which repairs work?

Exhibit 1: The first window, its simulation and the through case

	Value
First window, 504 returns (in-sample)	
Tangency weights: 0050 / 00646 / 00679B / 00635U / 00642U	11.8 / 62.6 / 37.8 / 2.6 / −14.7%
Best Sharpe ratio SR^*; equal weights' Sharpe ratio	0.880; 0.327
Experiment 1, 10,000 samples (model)	
Mean true Sharpe ratio: means and covariances estimated	0.423
Mean true Sharpe ratio: means estimated, covariances known	0.425
Mean true Sharpe ratio: covariances estimated, means known	0.87686
SD of the 00646 exposure: both estimated / covariances only (points)	67.1 / 4.73
Through case, 2019-03-05 → 2026-10-08 (out-of-sample)	
Realized volatility, net of costs: R1 / R3 / R5	12.48 / 9.96 / 9.97%

Note: E5's first window, 13 February 2017 to 27 February 2019; $A = 242.95$, $r_f = 1.5\%$. Weights are rounded separately; they sum to 100.1. SEs 0.003, 0.003 and 0.00002 for the Sharpe ratios, 0.5 and 0.03 points for the SDs. Fixed equal weights over the whole estimation frame had 11.49% (LM27). The realized volatilities are the standard deviations (divisor T) of the 1,851 net daily returns of the wealth frame, annualized with $A_{\mathrm{w}} = 243.57$.

Source: FinMind `TaiwanStockPrice`, `TaiwanStockDividendResult` and `TaiwanStockSplitPrice`, retrieved 24 October 2026 (`data/vol5_series.py`). Computations: `data/lm30_numbers.py`, `data/vol5_rules.py`.

Section 2 measures the errors, Section 3 sets the benchmark, Sections 4 to 7 test the repairs and Section 8 collects the Python.

> **NOTE** **Notation for this module**
>
> The first window is LM28's: $T = 504$ joint daily total returns of the five funds, $A = 242.95$. In Sections 2 and 3 its mean and $\hat{\Sigma}$ (divisor T) are the model's true $\boldsymbol{\mu}$ and Σ; $\mathbf{e} = \boldsymbol{\mu} - (r_f/A)\mathbf{1}$ with an illustrative $r_f = 1.5\%$; at 0% and 2% the model's SR^* would be 1.10 and 0.82 (LM28). These inputs are daily, unlike LM28's annual ones; δ, weights and Sharpe ratios do not depend on the unit. Years of sessions are $\tau = T/A$. U28 is LM11's: 28 sector price indices, $A = 245.4794$, halves of 2,191 returns with matrices $\hat{\Sigma}^{(1)}$, $\hat{\Sigma}^{(2)}$. A U28 portfolio's realized volatility is $\sqrt{A\,\mathbf{w}^\top \hat{\Sigma}^{(2)}\mathbf{w}}$, for weights reset every session.
>
> Local: $\hat{\mathbf{e}}$ excess sample means; $\hat{\mathbf{f}} = \hat{\Sigma}^{-1}\hat{\mathbf{e}}/\delta$ plug-in exposures at the section's δ; $\mathrm{SR}(\mathbf{f})$ a model Sharpe ratio; $\bar{\lambda} = \operatorname{tr}\hat{\Sigma}/N$; Σ_{tgt} a target; $\mathbf{q}$ Section 4's eigenvector of Σ; c the data's weight, or in (30.5) the view's; $\hat{c}$, c^*, κ_c (30.3)'s estimate and (30.2)'s oracle of (30.1)'s weight, and $\hat{\Sigma}_c$'s condition number; $\|M\|_F$ the Frobenius norm (LM18); $\sigma^2_{\hat{\Sigma}}$, $\hat{\sigma}^2_{\hat{\Sigma}}$ the noise of $\hat{\Sigma}$ and its estimate; ν_0, $\boldsymbol{\nu}$, $\hat{\Sigma}_\nu$, $\mathbf{w}^{\mathrm{lo}}$, $\mathbf{y}$ Section 5's; b, ε, $\mathbf{e}_{\mathrm{BL}}$, $\mathbf{f}_{\mathrm{BL}}$ Section 6's; $\hat{\Sigma}_{\mathrm{F}}$, $\hat{\beta}_i$, σ^2_{mkt}, $\hat{\sigma}^2_{\mathrm{res},i}$ Section 7's. Multipliers are ν, as λ_k and $\bar{\lambda}$ are eigenvalues; τ is years, so Black–Litterman's scalar is $1/n_0$. Kurtosis is raw, (14.5). Counts: k eigenvalues, i, j assets, t dates.

> **NOTE** **Research hygiene**
>
> The simulations of experiments 1 to 3 are **model** results: they draw from a fixed model built on real estimates. Weights, $\hat{c}$, resampled weights and Black–Litterman portfolios fitted to one window are **in-sample**. Two comparisons are **out-of-sample**: the through case's R1, R3 and R5 from March 2019, fixed in advance, and U28's estimators fitted on its first half and judged on the second.
>
> That split compares five estimators, and seven window lengths for two of them on a grid chosen after the halves' result; none is called the best. On E5 the module compares three of the through case's eight rules, which LM33 counts.

The weights fitted to one window in Example 6 to Example 8 and Exhibit 7 are plug-in estimates with an unmeasured error. Experiments 1, 3 and 4 measure such errors by simulation and resampling.

2 HOW FAR PLUG-IN WEIGHTS FALL FROM THE TRUTH

- [ ] simulate plug-in portfolios from a known model and measure the loss due to estimated means

A simulation can separate the errors, because it knows the truth. Take the first window's estimates as the true $\boldsymbol{\mu}$ and Σ of iid normal daily returns. Their annual means are 7.5, 10.0, 2.6, 0.8 and −1.4%, the

volatilities 13.5, 11.9, 8.5, 9.4 and 23.2%, and the correlations run from −0.30 to 0.54 (LM28). Samples colour standard normals with Σ's Cholesky factor (LM12).

Chopra and Ziemba (1993) found errors in means far more damaging to optimal portfolios than errors in variances or covariances, as LM7 reported. Experiment 1 measures the split.

With a risk-free asset, (28.8) gives the best exposures $\Sigma^{-1}\mathbf{e}/\delta$. At $\delta = \mathbf{1}^\top\Sigma^{-1}\mathbf{e} = 11.27$ they are the tangency weights of Exhibit 1, with $\mathrm{SR}^* = 0.880$. A sample gives three plug-ins at that fixed δ:

$$\hat{\mathbf{f}} = \frac{\hat{\Sigma}^{-1}\hat{\mathbf{e}}}{\delta}\ (\text{both estimated}),\quad \frac{\Sigma^{-1}\hat{\mathbf{e}}}{\delta}\ (\text{means only}),\quad \frac{\hat{\Sigma}^{-1}\mathbf{e}}{\delta}\ (\text{covariances only}).$$

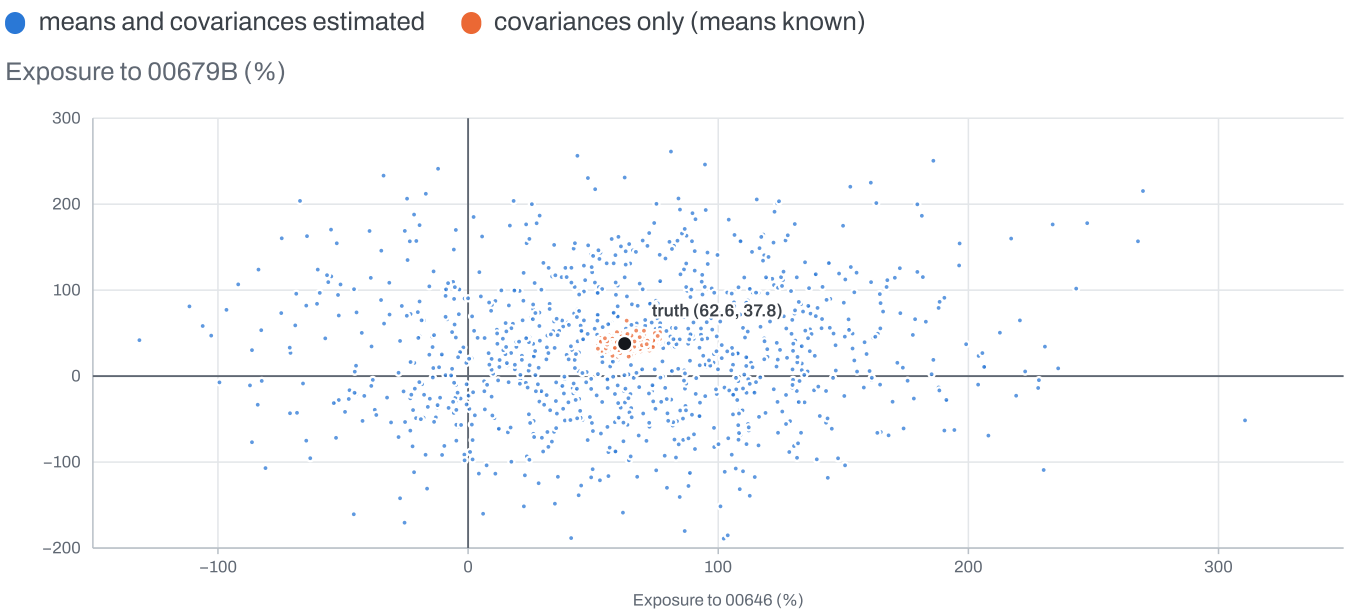
Each is judged by its true Sharpe ratio, with the model's $\mathbf{e}$ and Σ:

$$\mathrm{SR}(\mathbf{f}) = \frac{\sqrt{A}\,\mathbf{f}^\top\mathbf{e}}{\sqrt{\mathbf{f}^\top\Sigma\mathbf{f}}}.$$

By (16.4), $\mathbf{f}^\top\Sigma\mathbf{f}$ is the true variance of a portfolio built from estimates. A Sharpe ratio estimated from returns is itself noisy ((20.9)); LM33 tests the rules' Sharpe ratios, while here the model knows the true one.

The comparison uses exposures, not weights that sum to 1. In 2,115 of 10,000 samples (21%, SE 0.4 points) the sum $\mathbf{1}^\top\hat{\mathbf{f}}$ is zero or negative, so normalized weights would flip sign. Exhibit 2 shows the first 1,000 samples.

Exhibit 2: Plug-in exposures to 00646 and 00679B in experiment 1



Note: model results on E5's first window, 1,000 of the 10,000 samples. Exposures are scaled by the model's δ, so the truth's sum to 1. Over all samples the SDs are 67.1 and 80.7 points, against 4.73 and 4.79 with the covariances alone (batch SEs 0.5, 0.6, 0.03 and 0.04). Computations: `data/lm30_charts.py`.

With Σ known, (28.9) puts the means-only exposures at $\mathbf{w}^{\tan} + \Sigma^{-1}(\hat{\mathbf{e}} - \mathbf{e})/\delta$, and $\hat{\mathbf{e}} - \mathbf{e} \sim \mathcal{N}_5(\mathbf{0}, \Sigma/T)$. So by (16.4) the means-only exposures have

$$\operatorname{Cov}\left(\frac{\Sigma^{-1}\hat{\mathbf{e}}}{\delta}\right) = \frac{\Sigma^{-1}\Sigma\,\Sigma^{-1}}{\delta^2 T} = \frac{\Sigma^{-1}}{\delta^2 T}.$$

For 00646 this covariance gives an SD of 65.7 points, almost all of the spread in Exhibit 2.

EXAMPLE 1

Experiment 1: which errors cost

Draw 10,000 samples of 504 days from the model (`default_rng([1, 20261011])` , one `standard_normal((504, 5))` call per sample). Compare the three plug-ins. SEs of medians and SDs come from 20 batches of 500 samples.

SOLUTION

The mean true Sharpe ratio is 0.423 (SE 0.003) with both inputs estimated and 0.425 (SE 0.003) with the means alone. With the covariances alone it is 0.87686 (SE 0.00002). With both estimated the median is 0.489 (batch SE 0.004): a long left tail lowers the mean.

Exposure SDs run from 28.8 to 80.7 points with both estimated and from 1.75 to 4.79 with the covariances alone (batch SEs 0.2 to 0.6 and 0.01 to 0.04). The ratio runs from 14.2 (00646: 67.1 against 4.73) to 18.0 (00635U: 72.0 against 3.99), batch SEs 0.1.

The same draws measure the error in LM27's risk shares. R1's true shares, 1.1% to 50.2%, are estimated with SDs of 1.031 to 1.67 points (batch SEs 0.007 and 0.01).

EXAMPLE 2

Why the means

Compare the error of the first window's annual means with that of its covariance matrix.

SOLUTION

By (20.2) an annual mean's SE is $\sqrt{A}\,s/\sqrt{\tau}$, with $\tau = T/A = 2.0745$ years of sessions. The volatilities $\sqrt{A}\,s$, 13.50, 11.87, 8.54, 9.39 and 23.17%, give SEs of 9.4, 8.2, 5.9, 6.5 and 16.1 points. So four of the five means lie within one SE of zero.

By (18.4), for normal returns with a known mean, the root-mean-square error of $\hat{\Sigma}$ is 8.5% of $\|\Sigma\|_F$. More days over the same span shrink this error, not the means': (20.2) depends on years, (18.4) on days.

The plug-in minimum variance needs no mean. It predicts 4.990% (SE 0.002) and delivers 5.0353% (SE 0.0001), against the possible 5.0152%. By (28.3) its true variance exceeds $\sigma^2_{\min}$ by its weight error's quadratic form.

PITFALL **A fixed model measures sampling error alone**

The model's truth is an in-sample estimate: $\mathrm{SR}^* = 0.880$ is the first window's own optimum. Its returns are iid normal, and its means stay fixed, while real means move (LM24: the sector means of a later half spread almost twice as widely). The model is a best case; R1's and R3's realized volatilities are the evidence.

導讀 **平均報酬的誤差最傷**

把估計值（estimate）當成真值直接代入最佳化（optimization）公式，就是代入原則（plug-in principle）的做法。實驗一假設第一個視窗的估計值就是真值，再從這個模型抽出 504 天：同時估計平均與共變異數（covariance）時，真實夏普比率（Sharpe ratio）平均只剩 0.423；只估計共變異數時則有 0.87686，幾乎沒有損失。原因在於平均超額報酬（excess return）的估計誤差（estimation error）只隨年數下降：兩年資料讓 00646 年化平均的標準誤（standard error）達 8.2 個百分點，共變異數矩陣（covariance matrix）的誤差卻隨天數下降。這個模擬只量到抽樣誤差（sampling error）；真實的平均報酬還會移動，所以實際情況只會更糟。

KNOWLEDGE CHECK 1

How does the SE of 00646's annual mean change if the same two years held twice as many days? If the span doubled instead?

ANSWER

By (20.2) the SE is $\sqrt{A}\,s/\sqrt{\tau}$, and $\sqrt{A}\,s$ does not depend on how often returns are sampled. Twice the days over the same span leave the SE at 8.2 points; twice the span cuts it to $8.2/\sqrt{2} = 5.8$ points.

3 EQUAL WEIGHTS AS A BENCHMARK

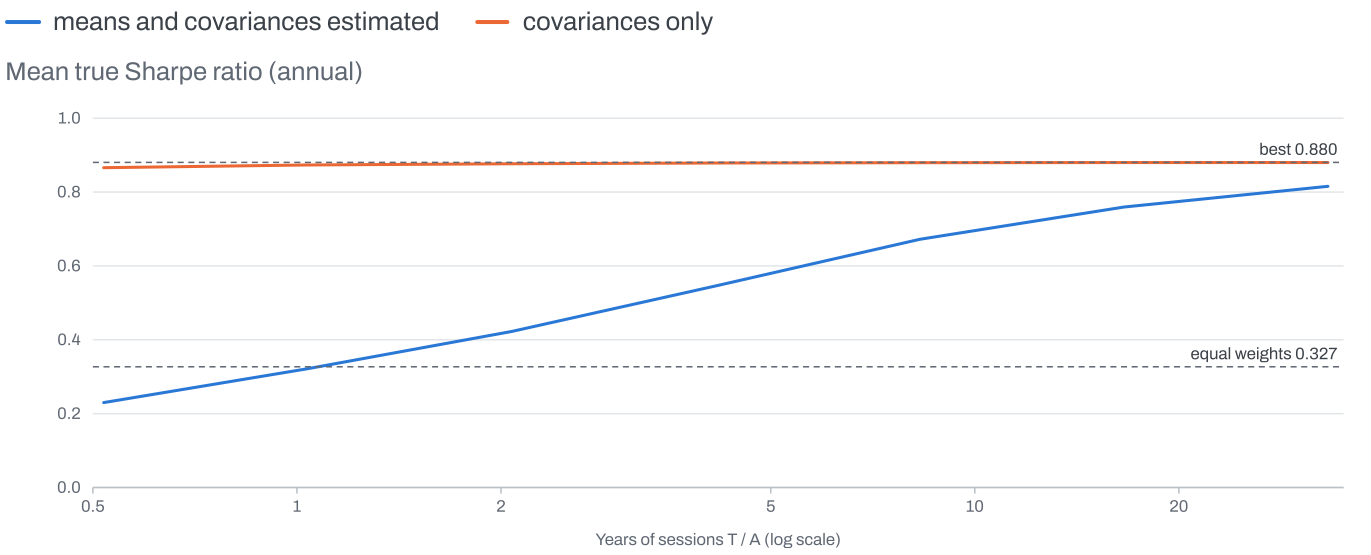
- [ ] explain why equal weights are hard to beat, and state the conditions of the 1/N evidence

Equal weights estimate nothing. By (10.6) the variance of the 1/N portfolio is $\overline{\text{var}}/N + (1 - 1/N)\,\overline{\text{cov}}$, so its risk depends only on the average variance and covariance. Its weights carry no estimation error, but they ignore every difference between the assets.

DeMiguel, Garlappi and Uppal (2009) compared 14 models with 1/N on seven datasets. None was consistently better in Sharpe ratio, certainty-equivalent return or turnover (their abstract). They name three conditions under which optimizing wins: a long estimation window, an ex ante best Sharpe ratio well above 1/N's, and few assets (p. 1919).

Experiment 2 asks how long a window the plug-in needs. It draws 2,000 samples at each of seven lengths from 126 to 8,064 days (`default_rng([2, 20261011])`), plotted in Exhibit 3.

Exhibit 3: Mean true Sharpe ratio of the plug-in portfolio by sample length



Note: model results of experiment 2 on E5's first window, 2,000 samples per length. Years of sessions are sessions over A, from 0.52 to 33.2. SEs 0.008 to 0.001 for the first series; 0.322 at 1.04 years of sessions lies within one SE of 1/N's 0.327. Computations: `data/lm30_charts.py`.

EXAMPLE 3

The critical window

When does the plug-in beat 1/N's true Sharpe ratio of 0.327 on average, and how does that compare with the US calibration?

SOLUTION

With both inputs estimated the mean true Sharpe ratio is 0.230, 0.322, 0.422, 0.546, 0.672, 0.759 and 0.816 at 0.52 to 33.2 years of sessions (SEs 0.008 to 0.001). With the covariances alone it is already 0.8660 (SE 0.0002) at 0.52 years of sessions. The plug-in is about equal to 1/N at 1.04 years of sessions (0.322, SE 0.008) and well above at 2.07 (0.422, SE 0.007).

DeMiguel, Garlappi and Uppal judge the sample-based rule in their §4 by its certainty-equivalent return instead. In their Figure 1's panel E, calibrated to US stocks, 1/N's monthly Sharpe ratio is 0.12 against the tangency's 0.15. The rule then needs more than 3,000 months with 25 assets to beat 1/N (pp. 1939–1941).

In their calibration 1/N reaches 0.8 of the best; in this model only 0.37 (0.327 against 0.880). The plug-in has much to gain, and the means do not move: a best case.

Kan and Zhou (2007) add that an optimal mix of the risk-free asset, the sample tangency portfolio and the sample minimum-variance portfolio dominates a mix of the first two alone.

On E5, out of sample and net of costs, R1 realized 12.48% and R3 9.96% on the 1,851 daily returns from 5 March 2019 to 8 October 2026.

R3's windows had predicted 8.84% on average, within LM28's range of 4.97% to 10.42%. An optimized in-sample variance is biased down, but in experiment 1 the bias is small: 4.990% predicted against 5.0353% delivered (Example 2). Most of the 1.1-point gap comes from inputs that moved after each window, as in LM7.

Costs, turnover and net growth of the rules are LM34's numbers, and the standard errors of their statistics LM33's; this module reports only volatility. Equal weights are also the $c = 0$ end of Section 4's shrinkage.

KNOWLEDGE CHECK 2

In experiment 1, 1/N has a true Sharpe ratio of 0.33 against the best 0.88, yet it beats the plug-in in 31.4% of samples (SE 0.5 points). Why can the worse portfolio win so often?

ANSWER

1/N's ratio is fixed, while the plug-in's varies. Errors in the means often point the exposures in a poor direction, so its ratio has a long left tail (mean 0.423, median 0.489). It falls below 0.327 in 31.4% of the samples.

4 SHRINKING THE COVARIANCE MATRIX

- ☐ derive the oracle shrinkage weight as a bias–variance trade-off
- ☐ compute Ledoit–Wolf shrinkage by hand and with scikit-learn, and diagnose its effect on eigenvalues and weights
- ☐ compare out-of-sample volatilities of sample, shrunk and constrained portfolios, labelling each result

Minimum-variance weights divide by the small eigenvalues of $\hat{\Sigma}$, (11.8), and the smallest is too small on average. With a known mean, (11.4) gives $\lambda_N(\hat{\Sigma}) \leq \mathbf{q}^\top \hat{\Sigma} \mathbf{q}$ for a unit eigenvector $\mathbf{q}$ of Σ's smallest eigenvalue, and $\mathbb{E}[\mathbf{q}^\top \hat{\Sigma} \mathbf{q}] = \lambda_N(\Sigma)$, so $\mathbb{E}\lambda_N(\hat{\Sigma}) \leq \lambda_N(\Sigma)$.

DEFINITION **Shrinkage toward the scaled identity**

For a weight $0 \le c \le 1$ on the data, with $\bar{\lambda} = \operatorname{tr}\hat{\Sigma}/N$,

$$\begin{aligned} \hat{\Sigma}_c &:= c\,\hat{\Sigma} + (1-c)\,\bar{\lambda} I, \\ \hat{\Sigma}_c \mathbf{q}_k &= \big(c\lambda_k + (1-c)\bar{\lambda}\big)\mathbf{q}_k, \\ \kappa_c &= \frac{c\lambda_1 + (1-c)\bar{\lambda}}{c\lambda_N + (1-c)\bar{\lambda}}. \end{aligned} \tag{30.1}$$

The target $\hat{\Sigma}_{\text{tgt}} = \bar{\lambda} I$ has the trace of $\hat{\Sigma}$.

By (11.3), shrinkage keeps the eigenvectors and moves every eigenvalue toward $\bar{\lambda}$, the small ones up and the large ones down, so κ falls. For $c < 1$ each eigenvalue is at least $(1-c)\bar{\lambda} > 0$: $\hat{\Sigma}_c$ is positive definite even when $\hat{\Sigma}$ is singular. At $c = 0$ the minimum-variance portfolio is 1/N. Unstable eigenvectors of close eigenvalues, (11.7), stay unrepaired.

THEOREM **The oracle weight**

Let Σ_{tgt} be a nonrandom matrix, and let $\hat{\Sigma}$ satisfy two conditions:

1. it is unbiased, $\mathbb{E}\hat{\Sigma} = \Sigma$ (a known mean);
2. its noise $\sigma^2_{\hat{\Sigma}} := \mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2$ is finite (finite fourth moments).

If $\|\Sigma - \Sigma_{\text{tgt}}\|_F^2 + \sigma^2_{\hat{\Sigma}} > 0$, then over nonrandom c the expected squared Frobenius distance of $c\hat{\Sigma} + (1-c)\Sigma_{\text{tgt}}$ from Σ is smallest at

$$c^* = \frac{\|\Sigma - \Sigma_{\text{tgt}}\|_F^2}{\|\Sigma - \Sigma_{\text{tgt}}\|_F^2 + \sigma^2_{\hat{\Sigma}}}. \tag{30.2}$$

For iid normal returns with a known mean, (18.4) gives $\sigma^2_{\hat{\Sigma}} = (\|\Sigma\|_F^2 + (\operatorname{tr}\Sigma)^2)/T$.

Proof. The error is $c(\hat{\Sigma} - \Sigma) + (1-c)(\Sigma_{\text{tgt}} - \Sigma)$, and its squared Frobenius norm is a sum of squared entries. The cross term has expectation zero, because $\mathbb{E}(\hat{\Sigma} - \Sigma) = 0$ and the second matrix is nonrandom. Hence

$$\begin{aligned} &\mathbb{E}\big\|c\hat{\Sigma} + (1-c)\Sigma_{\text{tgt}} - \Sigma\big\|_F^2 \\ &\quad = c^2\sigma^2_{\hat{\Sigma}} + (1-c)^2\|\Sigma - \Sigma_{\text{tgt}}\|_F^2, \end{aligned}$$

variance plus squared bias, (20.1)'s split entry by entry. This parabola in c is smallest where its derivative $2c\sigma^2_{\hat{\Sigma}} - 2(1-c)\|\Sigma - \Sigma_{\text{tgt}}\|_F^2$ is zero, at (30.2).

The oracle has the shape of LM24's (24.3), with the target's squared distance from the truth in the prior variance's place.

The oracle needs Σ. Ledoit and Wolf (2004a) estimate its parts:

$$\hat{c} = 1 - \frac{\min\left(\hat{\sigma}^2_{\hat{\Sigma}},\ \|\hat{\Sigma} - \bar{\lambda} I\|_F^2\right)}{\|\hat{\Sigma} - \bar{\lambda} I\|_F^2},$$
$$\hat{\sigma}^2_{\hat{\Sigma}} = \frac{1}{T^2} \sum_{t=1}^{T} \left\| (\mathbf{R}_t - \bar{\mathbf{R}})(\mathbf{R}_t - \bar{\mathbf{R}})^\top - \hat{\Sigma} \right\|_F^2. \tag{30.3}$$

The denominator estimates the oracle's, and $1 - \hat{c}$ estimates the noise's share $1 - c^*$ (Deep Dive). As N and T grow with N/T bounded, $\hat{c}$ is consistent and the shrunk matrix attains the oracle's loss asymptotically, under moment conditions. This is stated without proof (Theorems 3.2 and 3.4 of their 2001 working paper). Here N/T runs from 5/504 on E5 to 28/20 on U28.

Forecasting the matrix day by day, with the recursion (26.2) applied to $\mathbf{R}_t \mathbf{R}_t^\top$, is LM31's method.

EXAMPLE 4

Ledoit–Wolf by hand on the first window

Compute $\hat{c}$ for the first window by hand, check it with scikit-learn, and find where the noise comes from.

SOLUTION

The mean eigenvalue is $\bar{\lambda} = 0.0204$ a year, a volatility of 14.3%. With $\|\hat{\Sigma} - \bar{\lambda} I\|_F^2 = 1.885 \times 10^{-3}$ and $\hat{\sigma}^2_{\hat{\Sigma}} = 6.12 \times 10^{-5}$,

$$\hat{c} = 1 - \frac{6.12 \times 10^{-5}}{1.885 \times 10^{-3}} = 0.9675.$$

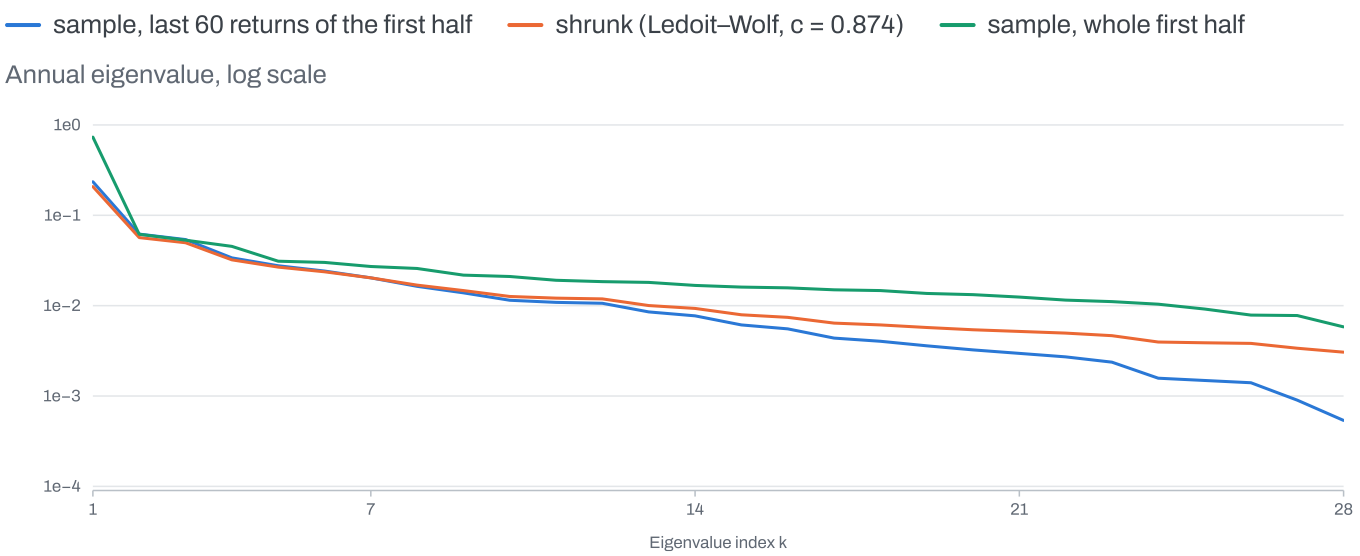
In scikit-learn `shrinkage_` is 0.0325, equal to $1 - \hat{c}$ to rounding error (below 10^{-15}).

With (18.4)'s normal value as the noise, the oracle is $c^* = 0.9851$; the noise estimate is 2.14 times that value. Fat tails cause the excess.

Entry by entry, (18.4)'s normal value is $(\hat{\sigma}_{ij}^2 + \hat{\sigma}_{ii}\hat{\sigma}_{jj})/T$, with $\hat{\sigma}_{ij}$ the entries of $\hat{\Sigma}$. A fund's share of the excess is its row's sum over the total. 0050's tails are the fattest (raw kurtosis 13.3), and its row carries 36%. The milder tails of 00642U (raw kurtosis 4.5) meet the largest variance (volatility 23.2%), so its row carries the most, 39%.

Exhibit 4 shows (30.1) on a 60-return window of U28.

Exhibit 4: U28's eigenvalues, sample and shrunk



Note: annual eigenvalues, largest first; log scale. The condition number is 438 for the last 60 returns of the first half, 68.1 after shrinkage ($\hat{c} = 0.874$) and 126.05 for the whole first half (LM11). Price indices, no dividends; in-sample. Source: FinMind `TaiwanStockPrice`, sector indices, retrieved 11 October 2026, closes repaired from `TaiwanStockEvery5SecondsIndex` as in LM10. Computations: `data/lm30_charts.py`.

Exhibit 5: Minimum variance on short windows of U28

T	$\hat{c}$	κ, sample → shrunk	Sample, predicted / realized (%)	Shrunk, realized (%)
20	0.790	singular → 49.5	—	12.59
30	0.823	116,097 → 59.9	1.34 / 43.12	12.46
60	0.874	438 → 68.1	3.96 / 13.85	12.04
120	0.894	127 → 49.7	4.49 / 11.15	11.11
250	0.941	92 → 59.3	5.53 / 11.10	11.17
504	0.981	92 → 80.8	6.79 / 11.49	11.51
2,191	0.995	126 → 121.4	11.40 / 11.78	11.78

Note: fitted on the last T returns of the first half (to 6 October 2017), judged on the second half. At $T = 20$ the sample matrix has rank 19; its pseudoinverse portfolio predicts 1.78% and realizes 17.03%. Price indices, no dividends. Realized volatilities are out-of-sample, one split per row; $\hat{c}$, κ and predictions are in-sample. The grid was chosen after the halves' result: an illustration, not evidence. Source: as Exhibit 4.

EXAMPLE 5

Few returns for many sectors

Fit minimum variance, sample and shrunk, to the last 30 and 20 returns of U28's first half, and judge both on the second half (Exhibit 5).

SOLUTION

On 30 returns the sample matrix has $\kappa = 116{,}097$. Its minimum-variance portfolio predicts 1.34% and realizes 43.12%, with gross exposure $\|\mathbf{w}\|_1 = 13.48$. Shrunk with $\hat{c} = 0.823$, it realizes 12.46%.

On 20 returns the sample matrix is singular, of rank 19, the most (10.7) allows. The vector $\mathbf{1}$ has 18% of its length outside the column space. Scaled to sum to 1, that part is a portfolio with zero predicted variance, so the minimum is 0. The null space has dimension $28 - 19 = 9$, so the minimizer is not unique.

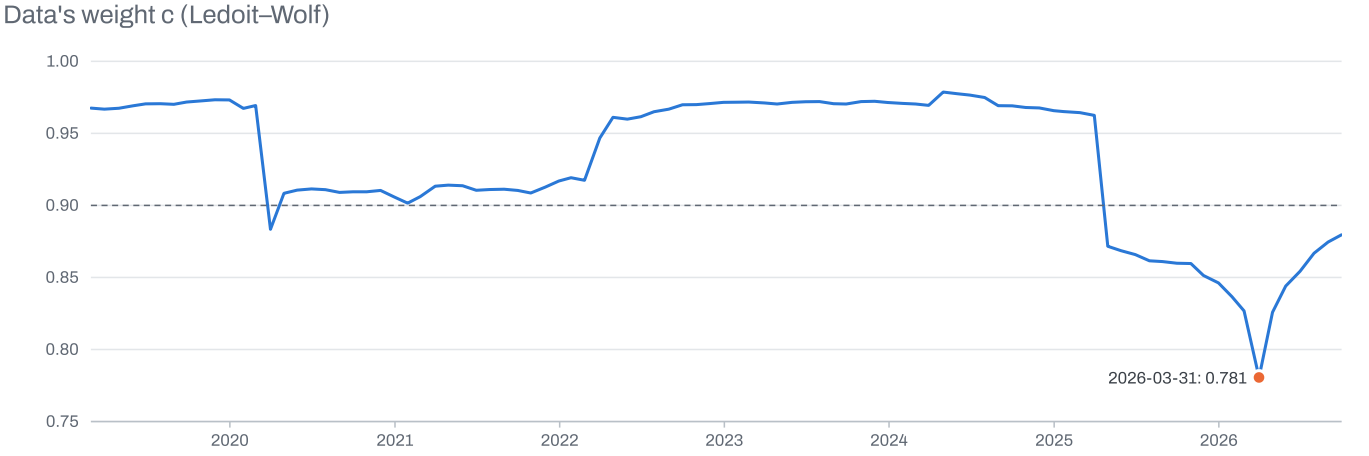
The pseudoinverse (12.7) gives the minimum within the column space, 1.78% predicted and 17.03% realized; the shrunk matrix realizes 12.59%.

On the whole first half ($\hat{c} = 0.9951$) κ moves from 126.05 to 121.38; both realize 11.78%.

Experiment 3 draws 1,000 samples at each length from $\hat{\Sigma}^{(1)}$, mean 0 (`default_rng([3, 20261011])`). At 60 returns the mean true volatility is 15.51% sample (SE 0.04) and 13.46% shrunk (SE 0.02); at 2,191, 11.4659% and 11.4651% (SEs 0.0006) against the possible 11.3952%. The mean $\hat{c}$ rises from 0.810 (SE 0.002) at 20 returns to 0.997963 (SE 0.000002) at 2,191.

Exhibit 6 shows $\hat{c}$ in R5's 92 windows of 504 returns.

Exhibit 6: Ledoit and Wolf's $\hat{c}$ at the through case's signals



Note: in-sample, since each $\hat{c}$ is fitted to its own window. First 0.968, median 0.962, highest 0.979 (2024-04-30), lowest 0.781 (2026-03-31); 19 signals lie below 0.9, the first on 2020-03-31. Source: as Exhibit 1. Computations: `data/lm30_charts.py`.

EXAMPLE 6

R5 against R3

R5 is R3 with the Ledoit–Wolf matrix. How far apart are their weights, and why did $\hat{c}$ fall in 2026?

SOLUTION

The largest gap between R5's and R3's weights is 13.1 points, at 2026-03-31, where $\hat{c}$ was lowest (00679B 55.1% $\rightarrow$ 42.0%). Over the 92 signals, the median of each signal's largest gap is 1.6 points.

In that window 00642U's raw kurtosis is 48.9, its row carries 84% of the excess noise, and the noise estimate is 13.1 times the normal value. With the normal value $\hat{c}$ would be 0.983, not 0.781: fat tails, not a short window, drove the shrinkage.

The target pulls each variance toward their average, a volatility of 26.5%: up for the bond fund (13.4%), down for 00642U (43.0%). So weight leaves the bond fund. Out of sample R5 realized 9.97% and R3 9.96%.

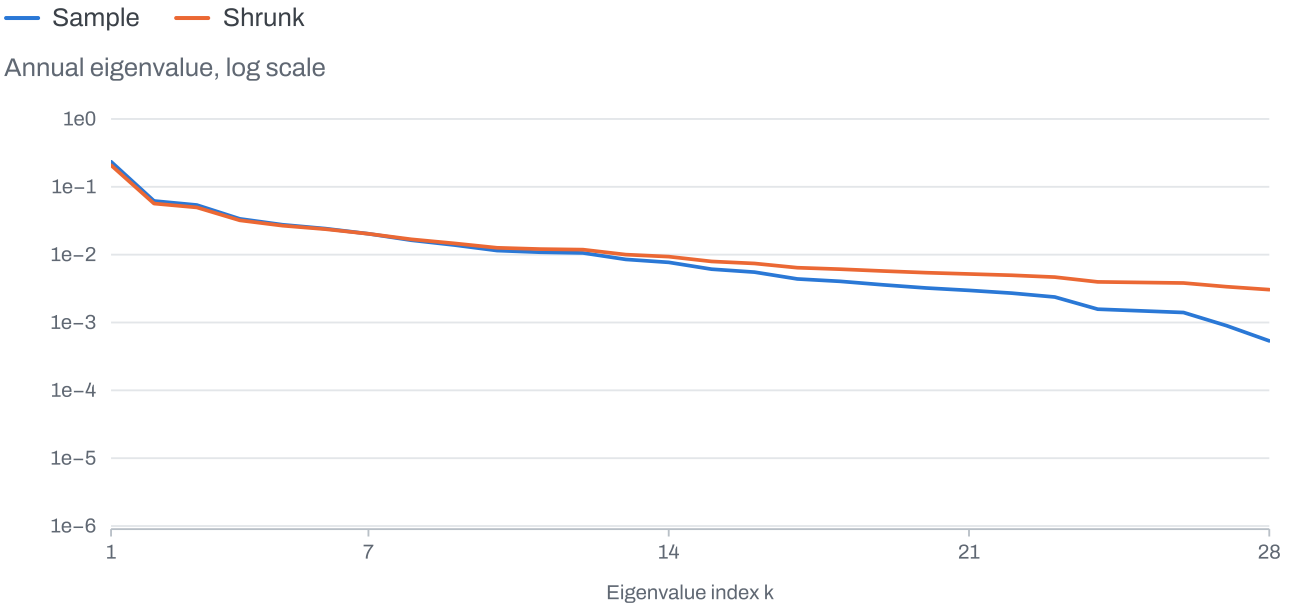
The widget below moves c by hand; its weights come from (11.8), with no optimizer:

$$\mathbf{w} \propto \sum_{k=1}^{N} \frac{\mathbf{q}_k^{\top}\mathbf{1}}{c\lambda_k + (1-c)\bar{\lambda}}\,\mathbf{q}_k.$$

On the last 60 returns at $\hat{c} = 0.874$ the widget shows $\kappa_c = 68.1$ and a realized 12.04%; at $c = 0$, 1/N's 15.51%.

Shrinkage from equal weights to minimum variance

INTERACTIVE



Ledoit–Wolf c (this window)
0.874

c used
0.874

Condition number κ_c
68.1

Predicted volatility
4.67%

Realized, second half
12.04%

Gross exposure (sum of |w|)
1.90

Largest weight
21.6%
CommunicationsInternet

Price indices, no dividends. The weights come from (11.8) with the shrunk eigenvalues of (30.1); no optimizer runs. Predicted volatility (in-sample) uses the shrunk matrix of the chosen window, realized volatility (out-of-sample) the second half (2017-10-11 to 2026-10-08), so the c that minimizes it is hindsight. At c = 0 the portfolio is 1/N.

DEEP DIVE **Deep Dive: why Ledoit and Wolf's noise estimate works** optional · click to expand

With a known mean, $\hat{\Sigma}$ is the average of T iid matrices $M_t = (\mathbf{R}_t - \boldsymbol{\mu})(\mathbf{R}_t - \boldsymbol{\mu})^\top$ with mean Σ. Entry by entry, (18.2) gives $\mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2 = \mathbb{E}\|M_t - \Sigma\|_F^2 / T$. In (30.3) the sum over T is the sample spread of the M_t, and the second $1/T$ is (18.2)'s.

The expansion in the proof of (30.2), applied to $(\hat{\Sigma} - \Sigma) + (\Sigma - \Sigma_{\text{tgt}})$, gives the Pythagorean identity of the 2001 working paper's §2,

$$\mathbb{E}\|\hat{\Sigma} - \Sigma_{\text{tgt}}\|_F^2 = \|\Sigma - \Sigma_{\text{tgt}}\|_F^2 + \sigma_{\hat{\Sigma}}^2.$$

So $\|\hat{\Sigma} - \bar{\lambda} I\|_F^2$ estimates the oracle's denominator, apart from the error in $\bar{\lambda}$; the minimum in (30.3) keeps $\hat{c} \geq 0$.

Stated without proof, the consistency result assumes (§3 of the working paper):

1. the ratio N/T stays bounded as both grow;
2. the returns have bounded eighth moments;
3. a condition holds on products of the returns' entries.

PITFALL **Ledoit and Wolf's estimate reacts to tails**

TR23's Student t fit has 3.065 degrees of freedom, so its fourth moment is infinite (LM15). Such returns make the noise of (30.2) infinite and fail Ledoit and Wolf's eighth-moment condition. In E5's window to 2026-03-31, $\hat{c}$ fell to 0.781 against 0.983 with the normal noise. Say what $\hat{c}$ responds to; never call it the right amount of shrinkage.

導讀 **收縮：用一點偏誤換掉大量雜訊**

平均已知時，樣本共變異數矩陣（sample covariance matrix）不偏（unbiased），但雜訊（noise）大；目標 $\bar{\lambda} I$ 沒有雜訊，卻有偏誤（bias）。收縮估計量（shrinkage estimator）取兩者的加權平均（weighted average），權重（weight）的選擇是偏誤及變異數之抵換（bias–variance tradeoff）：使 Frobenius 範數（Frobenius norm）下的均方誤差（mean squared error）最小。收縮（shrinkage）保留特徵向量（eigenvector），把每個特徵值（eigenvalue）往平均值拉，小的變大、大的變小，所以條件數（condition number）下降。28 個產業只用 60 筆報酬時，條件數從 438 降到 68.1，樣本外波動率從 13.85% 降到 12.04%；E5 用 504 天估 5 檔基金，收縮幾乎沒有影響。Ledoit–Wolf 的估計值會對厚尾（heavy tail）起反應，不能把它當成「正確的收縮量」。

KNOWLEDGE CHECK 3

A 2×2 correlation matrix with $\rho = 0.9$ is shrunk toward I with $c = 0.5$. What happens to its eigenvalues and condition number? Show that $c = 0$ gives the 1/N portfolio for any $\hat{\Sigma}$.

ANSWER

The eigenvalues $1 \pm \rho$ are 1.9 and 0.1, and $\bar{\lambda} = 1$. By (30.1) they become $0.5 \times 1.9 + 0.5 = 1.45$ and 0.55, so κ falls from 19 to $1.45/0.55 = 2.64$. At $c = 0$ the matrix is $\bar{\lambda} I$, whose inverse maps $\mathbf{1}$ to $\mathbf{1}/\bar{\lambda}$, so every weight is $1/N$.

5 CONSTRAINTS AS IMPLICIT SHRINKAGE

- [ ] derive the long-only minimum-variance portfolio as an unconstrained one on an adjusted covariance matrix

R3 forbids short positions. Jagannathan and Ma (2003) showed that this acts on the covariance matrix. The problem is

$$\min_{\mathbf{w}} \ \mathbf{w}^{\top} \hat{\Sigma} \mathbf{w} \quad \text{subject to} \quad \mathbf{1}^{\top} \mathbf{w} = 1, \ \ \mathbf{w} \geq \mathbf{0}.$$

The feasible set is closed and bounded, so a minimizer exists. With $\hat{\Sigma}$ positive definite the objective is strictly convex, so by (29.3) the minimizer $\mathbf{w}^{\text{lo}}$ is unique. LM29's KKT conditions (29.5), halved, read

$$\hat{\Sigma} \mathbf{w}^{\text{lo}} = \nu_0 \mathbf{1} + \boldsymbol{\nu}, \qquad \boldsymbol{\nu} \geq \mathbf{0}, \qquad \nu_i w_i^{\text{lo}} = 0.$$

Each multiplier is half of (29.5)'s, as Jagannathan and Ma scale them. Multiplying by $\mathbf{w}^{\text{lo}\top}$, with $\mathbf{1}^{\top} \mathbf{w}^{\text{lo}} = 1$ and $\boldsymbol{\nu}^{\top} \mathbf{w}^{\text{lo}} = 0$, gives $\nu_0 = s_p^2$. By (27.5), a held fund's sample covariance with the portfolio is s_p^2, an excluded one's $s_p^2 + \nu_i$.

PROPOSITION The constraint as an adjusted matrix

Let $\mathbf{w}^{\text{lo}}$ with $\mathbf{1}^{\top} \mathbf{w}^{\text{lo}} = 1$, ν_0 and $\boldsymbol{\nu}$ satisfy the halved KKT conditions above, and define

$$\hat{\Sigma}_{\nu} := \hat{\Sigma} - \left(\boldsymbol{\nu} \mathbf{1}^{\top} + \mathbf{1} \boldsymbol{\nu}^{\top} \right). \tag{30.4}$$

Then $\hat{\Sigma}_{\nu}$ is positive semidefinite and $\mathbf{w}^{\text{lo}}$ is one of its unconstrained minimum-variance portfolios (Jagannathan and Ma 2003, Proposition 1). If $\hat{\Sigma}$ is positive definite, so is $\hat{\Sigma}_{\nu}$, and $\mathbf{w}^{\text{lo}}$ is its only one.

Derivation. For any $\mathbf{x}$ put $\mathbf{y} = \mathbf{x} - (\mathbf{1}^{\top} \mathbf{x}) \mathbf{w}^{\text{lo}}$. Expanding $\mathbf{y}^{\top} \hat{\Sigma} \mathbf{y}$ with the KKT conditions and $\boldsymbol{\nu}^{\top} \mathbf{w}^{\text{lo}} = 0$ gives

$$\mathbf{x}^{\top} \hat{\Sigma}_{\nu} \mathbf{x} = \mathbf{y}^{\top} \hat{\Sigma} \mathbf{y} + \nu_0 \left(\mathbf{1}^{\top} \mathbf{x} \right)^2 \geq 0.$$

On the budget $\mathbf{1}^\top \mathbf{x} = 1$ the quadratic form is at least ν_0, the value at $\mathbf{w}^{\text{lo}}$, where $\mathbf{y} = \mathbf{0}$. With $\hat{\Sigma}$ positive definite and $\nu_0 > 0$, the sum is zero only if $\mathbf{y} = \mathbf{0}$ and $\mathbf{1}^\top \mathbf{x} = 0$, that is $\mathbf{x} = \mathbf{0}$. On the budget, equality needs $\mathbf{y} = \mathbf{0}$, that is $\mathbf{x} = \mathbf{w}^{\text{lo}}$. Jagannathan, Ma and Zhang (2019) correct the 2003 paper's proof of Proposition 2, not of Proposition 1.

The adjusted matrix lowers an excluded fund's covariances by $\nu_i + \nu_j$ and its variance by $2\nu_i$: the constraint shrinks only the entries of the funds it excludes.

EXAMPLE 7

The constraint on U28's first half

Solve the long-only problem on $\hat{\Sigma}^{(1)}$, build $\hat{\Sigma}_\nu$ and check (30.4).

SOLUTION

The smallest annual eigenvalue of $\hat{\Sigma}^{(1)}$ is 0.0058, so the minimizer is unique. It holds 9 sectors, so 19 constraints bind. The unconstrained minimum-variance weights of $\hat{\Sigma}_\nu$ equal the long-only weights to rounding error (below 10^{-15}). The smallest annual eigenvalue of $\hat{\Sigma}_\nu$ is 0.0059, and Optoelectronic's variance falls by 20.7%.

cvxpy's dual values give the same $\boldsymbol{\nu}$ to 3.7×10^{-10} of the largest ν_i. In the second half the long-only portfolio realized 12.14%, against 11.78% unconstrained.

In experiment 3, paired on the same samples, the constraint lowers the true volatility against shrinkage by 0.38 points (SE 0.03) at 30 returns and 0.06 (SE 0.02) at 60. It raises it by 0.49 points (SE 0.01) at 120 and 1.2981 (SE 0.0005) at 2,191: wrong constraints cost when data are long. R3's constraint binds at none of the through case's 92 signals (LM29).

導讀 **不可放空，等於修改共變異數矩陣**

加上非負性條件（nonnegativity condition）的最小變異數投資組合（minimum-variance portfolio），等於另一個矩陣 $\hat{\Sigma}_\nu$ 的無限制最小變異數解。被排除的資產與投資組合的共變異數（covariance）偏高，它的乘數（multiplier）ν_i 為正，這條限制條件（constraint）就是有效限制（active constraint）。調整後的矩陣把它的共變異數降低 $\nu_i + \nu_j$、變異數（variance）降低 $2\nu_i$，所以限制條件本身就是一種收縮（shrinkage）。資料少時它有幫助；資料多而真實的最適解需要放空時，它反而有成本。

KNOWLEDGE CHECK 4

On U28's first half 19 constraints bind. Which way did the constraint move those sectors' covariances, and why did the second half's volatility still rise, from 11.78% to 12.14%?

ANSWER

Down: $\hat{\Sigma}_\nu$ lowers each excluded sector's covariances by $\nu_i + \nu_j$ and its variance by $2\nu_i$, up to 20.7% for Optoelectronic. With 2,191 returns the sample matrix is precise, so this implicit shrinkage adds bias and removes little noise.

6 STARTING FROM A REFERENCE PORTFOLIO: BLACK–LITTERMAN

- [ ] derive the Black–Litterman posterior for one view and compute the weights it implies

Means are the weak input. The model of Black and Litterman (1992) starts from a reference portfolio and lets views move it. If an investor with risk aversion δ holds $\mathbf{w}_{\text{ref}}$ optimally, (28.8) gives $\mathbf{w}_{\text{ref}} = \hat{\Sigma}^{-1}\mathbf{e}/\delta$. Reversed, it gives the **implied excess returns**

$$\mathbf{e}_{\text{eq}} := \delta\,\hat{\Sigma}\mathbf{w}_{\text{ref}}.$$

The reference here is R2, LM27's *constructed* mix of 40% 0050, 20% 00646, 30% 00679B and 10% 00635U. The book has no market weights for these funds. The risk aversion

$$\delta = \frac{\mathbf{w}_{\text{ref}}^{\top}\hat{\mathbf{e}}}{\mathbf{w}_{\text{ref}}^{\top}\hat{\Sigma}\mathbf{w}_{\text{ref}}}$$

makes R2's implied mean excess return equal its window mean.

The prior is $\mathbf{e} \sim \mathcal{N}_5(\mathbf{e}_{\text{eq}}, \hat{\Sigma}/n_0)$: the reference is worth n_0 observations, as in LM24. The literature's scalar is $1/n_0$, with no consensus value. Idzorek (2007) reports values near 0, from 0.01 to 0.05, 1, and about one over the number of observations, each from other authors.

A view is a linear statement about excess returns, with an error: $K\mathbf{e} = b + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \Xi)$ is independent of $\mathbf{e}$. Here $K = (-1, 1, 0, 0, 0)$, so the view says that 00646 beats 0050 by b.

PROPOSITION **The Black–Litterman posterior for one view**

Under this prior and view, the posterior mean of $\mathbf{e}$ and its exposures $\mathbf{f}_{\mathrm{BL}} = \hat{\Sigma}^{-1}\mathbf{e}_{\mathrm{BL}}/\delta$ are

$$\begin{aligned} \mathbf{e}_{\mathrm{BL}} &= \mathbf{e}_{\mathrm{eq}} + \hat{\Sigma}K^{\top}\,\frac{c\,(b - K\mathbf{e}_{\mathrm{eq}})}{K\hat{\Sigma}K^{\top}}, \\ c &= \frac{K\hat{\Sigma}K^{\top}/n_0}{K\hat{\Sigma}K^{\top}/n_0 + \Xi}, \\ \mathbf{f}_{\mathrm{BL}} &= \mathbf{w}_{\mathrm{ref}} + K^{\top}\,\frac{c\,(b - K\mathbf{e}_{\mathrm{eq}})}{\delta\,K\hat{\Sigma}K^{\top}}. \end{aligned} \tag{30.5}$$

A data view, $\Xi = K\hat{\Sigma}K^{\top}/T$, gives $c = T/(T + n_0)$, (24.3)'s weight.

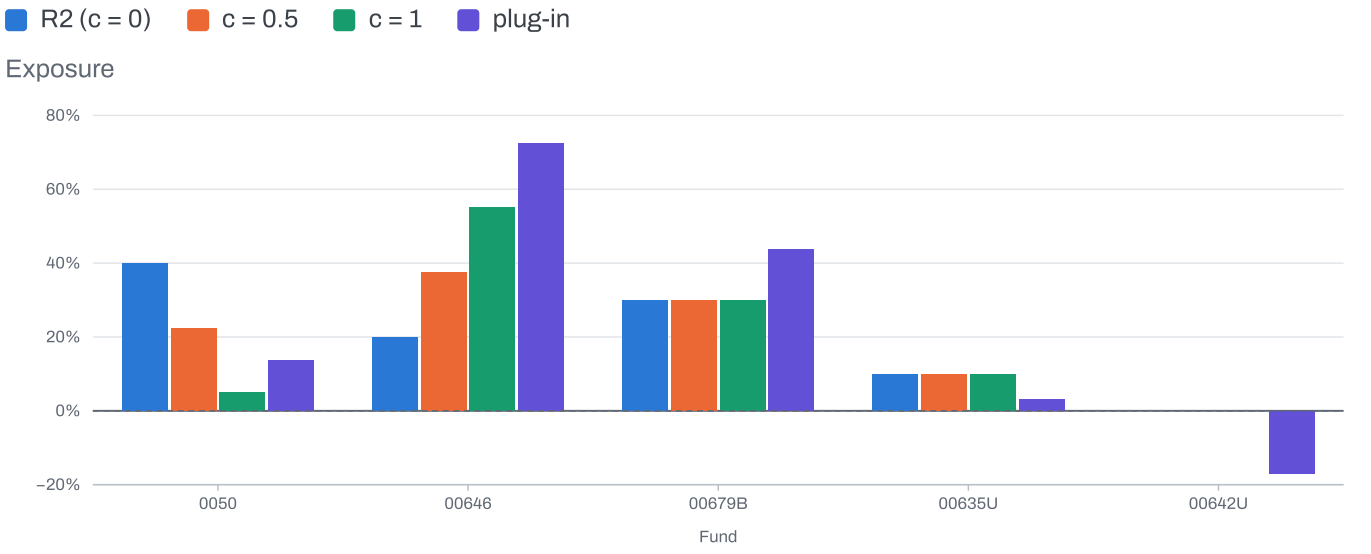
Derivation. By (16.4) and the independence of ε, each e_i and the observed view $K\mathbf{e} + \varepsilon$ form a normal pair, with covariance $(\hat{\Sigma}K^{\top})_i/n_0$ and view variance $K\hat{\Sigma}K^{\top}/n_0 + \Xi$. By (16.8), the mean of e_i given that the view equals b is the prior mean plus the covariance over the view variance times $b - K\mathbf{e}_{\mathrm{eq}}$: (30.5)'s first line. Multiplying by $\hat{\Sigma}^{-1}/\delta$, with $\hat{\Sigma}^{-1}\mathbf{e}_{\mathrm{eq}}/\delta = \mathbf{w}_{\mathrm{ref}}$, gives the last.

The exposures move along $K^{\top}$ only. He and Litterman (1999) state the general case: the unconstrained optimum is the scaled reference plus a weighted sum of view portfolios. The posterior mean for several views is stated without proof, in an equivalent form of Idzorek's (2007) formula (3). With one row of K per view and Ξ their error covariance,

$$\begin{aligned} \mathbf{e}_{\mathrm{BL}} = \mathbf{e}_{\mathrm{eq}} \\ + \hat{\Sigma}K^{\top}\left(K\hat{\Sigma}K^{\top} + n_0\Xi\right)^{-1}(\mathbf{b} - K\mathbf{e}_{\mathrm{eq}}). \end{aligned}$$

The exposures here use $\hat{\Sigma}$, not the posterior covariance of $\mathbf{e}$. Exhibit 7 traces them as c rises.

Exhibit 7: Black–Litterman exposures on the first window



Note: in-sample, first window; R2 is *constructed*; $\delta = 9.74$; one view, $K = (-1, 1, 0, 0, 0)$, with b the window's own difference. Source: as Exhibit 1. Computations: `data/lm30_charts.py`.

EXAMPLE 8

One view on the first window

Calibrate R2 on the first window, take the view that 00646 beats 0050, and find the exposures at $c = 0.5$ and $c = 1$.

SOLUTION

R2's mean excess return is 4.383% a year and its volatility 6.709%, so $\delta = 0.04383/0.06709^2 = 9.74$, and $\mathbf{e}_{\text{eq}}$ is 7.79, 5.14, 0.46, 1.02 and 3.71%.

The window says $b = 2.491$ points; the reference implies $K\mathbf{e}_{\text{eq}} = -2.649$ points, and $\sqrt{K\hat{\Sigma}K^{\top}} = 12.264\%$. With the view treated as a data view, $\Xi = K\hat{\Sigma}K^{\top}/504$, and $n_0 = 504$, the weight is $c = 0.5$, and (30.5) moves $0.5 \times 0.05140/(9.74 \times 0.12264^2) = 0.175$, or 17.5 points, from 0050 to 00646. So $\mathbf{f}_{\text{BL}}$ is 22.5 / 37.5 / 30 / 10 / 0%, and at $c = 1$ it is 4.9 / 55.1 / 30 / 10 / 0%.

The plug-in at the same δ holds 13.6 / 72.4 / 43.8 / 3.0 / −17.1%, a total of 115.8% before rounding.

Jorion (1986) argued that an empirical Bayes estimate of the means should outperform the sample mean in a portfolio. Sizing a position when the means are uncertain is LM45's subject.

PITFALL **The reference carries the prior**

Black–Litterman moves away from $\mathbf{w}_{\text{ref}}$ only as far as the views push, so a poor reference is a poor prior. R2 is *constructed*, and δ fits R2's own window mean; report both beside every such portfolio.

導讀 **從參考投資組合出發**

Black–Litterman 不直接採用樣本平均，而是先假設參考投資組合（reference portfolio）是最適的，反推出隱含超額報酬（implied excess returns）$\mathbf{e}_{\text{eq}} = \delta\hat{\Sigma}\mathbf{w}_{\text{ref}}$，作為事前分布（prior distribution）的中心。觀點（view）是對超額報酬的線性陳述，帶有誤差；事後分布（posterior distribution）的平均把兩者依精確度（precision）加權，與 LM24 的 (24.3) 同形。只有一個觀點時，部位只沿著觀點的方向移動，本例只有 0050 與 00646 改變。參考投資組合 R2 是建構的，事前分布的好壞取決於它。

KNOWLEDGE CHECK 5

Take all five window means as views ($K = I$, $\mathbf{b} = \hat{\mathbf{e}}$, $\Xi = \hat{\Sigma}/T$). Show that $\mathbf{f}_{\mathrm{BL}} = c\hat{\mathbf{f}} + (1-c)\mathbf{w}_{\mathrm{ref}}$ with $\hat{\mathbf{f}}$ at the same $\delta = 9.74$, and find 0050's exposure at $c = 0.5$ from the plug-in's 13.6%.

ANSWER

With $K = I$ the several-view formula gives

$$\begin{aligned}\mathbf{e}_{\mathrm{BL}} &= \mathbf{e}_{\mathrm{eq}} + \hat{\Sigma}\Big(\hat{\Sigma} + \frac{n_0}{T}\hat{\Sigma}\Big)^{-1}(\hat{\mathbf{e}} - \mathbf{e}_{\mathrm{eq}}) \\ &= c\,\hat{\mathbf{e}} + (1-c)\,\mathbf{e}_{\mathrm{eq}},\end{aligned}$$

with $c = T/(T+n_0)$. Multiplying by $\hat{\Sigma}^{-1}/\delta$ gives the result. For 0050, $0.5 \times 13.6\% + 0.5 \times 40\% = 26.8\%$.

7 RESAMPLING, FACTOR STRUCTURE AND REPAIR

- ☐ choose a repair (shrinkage, constraints, a reference portfolio, a factor model or resampling) for a given ratio of observations to assets
- ☐ compare out-of-sample volatilities of sample, shrunk and constrained portfolios, labelling each result

Michaud (1989) argued that mean–variance optimization tends to maximize the effects of errors in its inputs, and that unconstrained optimization can do worse than equal weights. Resampled efficiency, first proposed by Michaud (1998), averages instead, as Michaud and Michaud (2008) describe it. It draws means and covariances around the estimates, computes each draw's frontier, and averages the weights of matched portfolios.

Experiment 4 is a variant (`default_rng([4, 20261011])`). It averages R4's long-only maximum-Sharpe portfolio (29.6) over 1,000 iid bootstrap resamples of the first window's rows (LM22). In the 12 where no fund's mean exceeds r_f, R3's minimum variance is used instead. The average holds every fund, from 3.8% (00642U) to 35.2% (00646), SEs 0.4 and 0.9 points.

Yet each fund is excluded in 18% (00646) to 72% (00642U) of the resamples, SEs 1 to 2 points: averaging spreads weights that each resample concentrates.

A factor structure cuts the parameters. The single-index matrix of Sharpe (1963) is LM27's (27.8) with uncorrelated residuals,

$$\hat{\Sigma}_{\mathrm{F}} = \sigma^2_{\mathrm{mkt}}\,\hat{\boldsymbol{\beta}}\hat{\boldsymbol{\beta}}^{\top} + \operatorname{diag}\left(\hat{\sigma}^2_{\mathrm{res},1}, \ldots, \hat{\sigma}^2_{\mathrm{res},N}\right),$$

with (23.1)'s betas on the TAIEX price index and residual variances with divisor T. For U28 the single-index matrix has 57 parameters against the sample's 406, yet its portfolio realized 12.67% in the second

half, against the sample's 11.78% (Exhibit 8): a wrong structure costs. Ledoit and Wolf shrink toward this matrix (2003) or toward constant correlations (2004b); nonlinear shrinkage is outside this book.

Exhibit 8: Five estimators fitted on U28's first half

Estimator	Predicted (%)	Realized (%)
Sample	11.40	11.78
Shrunk (Ledoit–Wolf)	11.41	11.78
Long-only	12.74	12.14
Single index	11.01	12.67
1/N	15.98	15.51
Best c in hindsight, 0.98		11.78

Note: minimum-variance portfolios fitted on the first half (2,191 returns), judged on the second. Price indices, no dividends. Realized volatilities are out-of-sample on LM11's split, fixed in advance, and predictions in-sample. Five estimators are compared, and none is called the best. The last row is in-sample, chosen on the second half. Source: as Exhibit 4; the single index is the TAIEX price index (`TaiwanStockPrice`).

EXAMPLE 9

Repairing LM10's pairwise matrix

LM10's constructed C_{pw} has entries 0.982, 0.199 and −0.444 and $\det C_{\text{pw}} = -0.374592$. Repair it three ways.

SOLUTION

Its eigenvalues are 2.0149, 1.1471 and −0.1621. Shrinking toward I gives eigenvalues $c\lambda_k + 1 - c$, the smallest nonnegative up to $c = 1/(1-\lambda_3) = 0.8605$, with off-diagonal entries 0.845, 0.171 and −0.382. Clipping λ_3 to 0 and rescaling to a unit diagonal gives 0.852, 0.154 and −0.385.

The nearest correlation matrix (Higham 2002b) gives 0.864, 0.139 and −0.379, PSD to rounding; a second solver, cvxpy's CLARABEL, agrees to 1.4×10^{-7}. The Frobenius distances to C_{pw} are 0.216, 0.211 and 0.209.

The nearest matrix minimizes $\|M - C_{\text{pw}}\|_F^2$ over PSD matrices M with unit diagonal. That set is closed, bounded and convex, and the objective strictly convex, so the minimizer exists and is unique. Rousseeuw and Molenberghs (1993) compare methods based on shrinking, on eigenvalues and on geometry. Constructed (LM10's).

The Summary's table chooses a repair by the ratio T/N of observations to assets. Picking the best of several estimators judged on one split is a search, which LM34 corrects.

導讀 **重抽樣、因子結構與修補**

重抽樣（resampling）把資料重抽很多次，每次求最適權重再取平均；平均後每檔基金都有部位，但每次重抽都圍繞同一組樣本平均。單一指標模型（single index model）用市場的 beta 描述共變異數（covariance），28 個產業只要 57 個參數（parameter），但結構不對時樣本外波動率反而較高。成對估計的相關矩陣（correlation matrix）可能不是半正定（positive semidefinite）：可以往單位矩陣（identity matrix）收縮、把負的特徵值（eigenvalue）截為零，或找最近的相關矩陣。

KNOWLEDGE CHECK 6

Show that the largest c that keeps $cC_{\text{pw}} + (1-c)I$ PSD is $1/(1-\lambda_3)$, and evaluate it.

ANSWER

The matrix has C_{pw}'s eigenvectors and eigenvalues $c\lambda_k + 1 - c$. The smallest, $1 - c(1-\lambda_3)$, is nonnegative for $c \le 1/(1-\lambda_3)$. With $\lambda_3 = -0.162$ that is $1/1.162 = 0.861$.

8 ESTIMATION-ROBUST PORTFOLIOS IN PYTHON

- ☐ compute Ledoit–Wolf shrinkage by hand and with scikit-learn, and diagnose its effect on eigenvalues and weights

PYTHON

```python
import cvxpy as cp
import numpy as np
import polars as pl
from sklearn.covariance import LedoitWolf

est = pl.read_csv("data/e5_estimation.csv", schema_overrides={"fund": pl.Utf8})
wide = est.pivot(on="fund", index="date", values="R").sort("date")
funds = ["0050", "00646", "00679B", "00635U", "00642U"]
first = wide.filter(pl.col("date") <= pl.lit("2019-02-27")).tail(504)
x = first.select(funds).to_numpy()              # 504 x 5 daily returns
lw = LedoitWolf(assume_centered=False).fit(x)
c_hat = 1 - lw.shrinkage_                       # 0.9675
A = 242.951422                                  # E5's estimation frame
S = A * np.cov(x, rowvar=False, ddof=0)         # annual inputs for solvers
w = cp.Variable(5)
cons = [cp.sum(w) == 1, w >= 0]
prob = cp.Problem(cp.Minimize(cp.quad_form(w, cp.psd_wrap(S))), cons)
prob.solve(solver=cp.CLARABEL, tol_gap_abs=1e-12, tol_gap_rel=1e-12,
           tol_feas=1e-12)
assert prob.status == "optimal"
nu = cons[1].dual_value / (2 * A)               # Jagannathan and Ma's scale
```

`LedoitWolf(assume_centered=False)` removes the mean, divides by T and shrinks toward $\bar{\lambda} I$ only; `shrinkage_` is the target's weight $1 - \hat{c}$. `OAS` is another estimator, after Chen et al. (2010); its source says it omits their $2/p$ factor. The dual values of `w >= 0` belong to $\mathbf{w}^\top (A\hat{\Sigma})\mathbf{w}$, so dividing by $2A$ gives the daily $\boldsymbol{\nu}$. `psd_wrap` asserts positive semidefiniteness without testing it.

On the first window no constraint binds, so every ν_i is 0 to solver tolerance; Example 7 checks the scale on U28.

> **IN PYTHON** **Companion lab**
>
> The Polars-first notebook `LM30_lab.ipynb` rebuilds the four experiments, Ledoit–Wolf by hand and with scikit-learn, U28's windows and halves, and R5 against the through case's targets. It also rebuilds (30.4) with cvxpy's dual values, Black–Litterman and the repairs of C_{pw}. It solves the computational practice problems and ends with LM7's monthly re-estimation question, answered with shrinkage.
>
> The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 7

For the first window, scikit-learn reports `shrinkage_` 0.0325. What is $\hat{c}$, and what happens to it if the returns are annualized first?

ANSWER

The data's weight is $\hat{c} = 1 - 0.0325 = 0.9675$. Scaling the returns by $\sqrt{A}$ scales every matrix in (30.3) by A and both squared norms by A^2, so their ratio, and $\hat{c}$, do not change.

SUMMARY

- Plug-in weights inherit the means' errors: on E5's model the mean true Sharpe ratio is 0.423 of 0.880, but 0.87686 with the means known.
- A mean's error falls only with years, (20.2); a covariance matrix's falls with days, (18.4).
- Equal weights carry no estimation error; optimizing wins under three conditions: a long window, a large gap in Sharpe ratios and few assets.
- Shrinkage (30.1) pulls the eigenvalues toward $\bar{\lambda}$; the oracle (30.2) trades bias against noise, and (30.3) estimates it.
- The long-only constraint is minimum variance on the adjusted matrix (30.4).
- Black–Litterman (30.5) moves the reference along the views, with (24.3)'s weight for a data view.
- Resampling spreads weights, a factor model cuts parameters, and three repairs make a correlation matrix PSD.

Repair	What it fixes	When it helps (T/N)	Its cost
Shrinkage toward $\bar{\lambda}I$	noise and small eigenvalues of $\hat{\Sigma}$	small: 30 returns for 28 sectors	pulls variances together; reacts to tails
Long-only constraint	short positions built on noise	small: 30 to 60 returns in experiment 3	bias when the truth needs shorts
Reference portfolio	errors in means	any, given a credible reference	the reference is the prior
Single-index matrix	too many parameters	many assets (Sharpe 1963); not tested here at small T/N	a wrong structure (12.67% against 11.78% at $T/N \approx 78$)
Resampling	weights that jump	any; an illustration here	still centred on the window's means
Equal weights	every estimate	short windows, a small best Sharpe ratio	ignores all differences

PRACTICE PROBLEMS

Problems on data use Section 1's samples, divisor T and the target $\bar{\lambda} I$; multipliers are scaled as $\hat{\Sigma}\mathbf{w} = \nu_0 \mathbf{1} + \boldsymbol{\nu}$.

1. Daily returns have a volatility of $\sigma = 20\%$ a year, with $A = 243$, and a sample holds $T = 972$ returns, four years of sessions. The SE of the annual mean is *closest* to:

   A. 0.64 points

   B. 5.0 points

   C. 10.0 points

2. A 2×2 covariance matrix has eigenvalues 4 and 1 ($\%^2$ a day). Shrunk toward $\bar{\lambda} I$ with $c = 0.8$, its condition number is *closest* to:

   A. 2.85

   B. 4.00

   C. 3.20

3. Returns are iid normal with a known mean, $\Sigma = \sigma^2 I$, $N = 10$ and $T = 250$. The oracle weight c^* of (30.2) toward $(\operatorname{tr}\Sigma/N) I$ is:

   A. 0.96

   B. 0

   C. 1

4. A data view rests on $T = 504$ returns, and the prior is worth $n_0 = 1{,}512$ observations. The view says 4%, and the reference implies 0%. The posterior mean along the view is *closest* to:

   A. 1.0%

   B. 2.0%

   C. 3.0%

5. Two funds have $\sigma_1 = 20\%$ a year and correlation $\rho = 0.9$. Whatever the second fund's volatility, the $(1,1)$ entry of Σ^{-1} (a year) is *closest* to:

   A. 250.0

   B. 25.0

   C. 131.6

6. In a long-only minimum-variance problem, $w_3 \geq 0$ binds with $\nu_3 = 0.0002$ (daily units). In $\hat{\Sigma}_\nu$, asset 3's variance is lowered by:

   A. $\nu_3 = 0.0002$

   B. $2\nu_3 = 0.0004$

C. 0, since only covariances change

7. Derive (30.2). Evaluate it toward $(\operatorname{tr}\Sigma/N)I$ for $\Sigma = \operatorname{diag}(4, 1)$ (%² a day), normal returns with a known mean and $T = 100$, to four decimals.

8. Derive (30.4) from the KKT conditions. Two funds have daily volatilities of 1% and 2% and $\rho = 0.8$. Find the unconstrained and the long-only minimum-variance weights, ν_0 and ν_2 (%² a day), $\hat{\Sigma}_\nu$ and its minimum-variance weights.

9. Derive (30.5) for one view by (16.8). Two funds have volatilities of 20% and 10% a year and $\rho = 0$ (their matrix is $\hat{\Sigma}$), $\mathbf{w}_{\text{ref}} = (50\%, 50\%)$ and $\delta = 4$. A view $b = 0$ on fund 2 minus fund 1 has $c = 0.5$. Find $\mathbf{e}_{\text{eq}}$, $\mathbf{e}_{\text{BL}}$ and $\mathbf{f}_{\text{BL}}$.

10. *(Python)* On E5's first window, compute Ledoit–Wolf by hand and with scikit-learn, and the normal oracle. Find the ratio of the noise estimate to its normal value and each fund's share of the excess. What do they say about the tails?

11. *(Python)* Rerun experiment 1 (`default_rng([1, 20261011])`, 10,000 samples). Report the three mean true Sharpe ratios with SEs, and the count with $\mathbf{1}^\top\hat{\mathbf{f}} \le 0$. Also report the share of samples in which 1/N's true Sharpe ratio exceeds that of $\hat{\mathbf{f}}$, with both inputs estimated. Which of DeMiguel, Garlappi and Uppal's three conditions does the first window meet?

12. *(Python)* (a) Rebuild Exhibit 5's rows for 30 and 2,191 returns, label each result, and say which repair T/N favours at each length. (b) Check (30.4) on the first half. (c) Which repairs of the Summary's table suit 60 returns of 28 sectors, and 504 returns of five funds? Name each one's cost.

SOLUTIONS

1. **C** is correct: by (20.2) the SE is $\sigma/\sqrt{\tau} = 20/\sqrt{4} = 10.0$ points. A divides by the square root of the 972 days, B by the years instead of their square root.

2. **A** is correct: $\bar{\lambda} = 2.5$, and (30.1) gives $0.8 \times 4 + 0.2 \times 2.5 = 3.70$ and $0.8 \times 1 + 0.2 \times 2.5 = 1.30$, so $\kappa_c = 2.85$. B reads shrinkage as a rotation that keeps the eigenvalues. C multiplies $\kappa = 4$ by c.

3. **B** is correct: the target equals the truth, so $\|\Sigma - \Sigma_{\text{tgt}}\|_F^2 = 0$ and (30.2) puts all weight on the target. A borrows the form $T/(T+N)$ from (24.3). C would need a noiseless sample matrix.

4. **A** is correct: $c = 504/(504 + 1{,}512) = 0.25$, and (30.5) moves the view's value from 0% by $0.25 \times 4\% = 1.0\%$. C swaps c and $1 - c$. B weighs the two equally.

5. **C** is correct: by (9.5), $(\Sigma^{-1})_{11} = \sigma_2^2/\det\Sigma = 1/(\sigma_1^2(1-\rho^2))$, or VIF $/\sigma_1^2 = 5.263/0.04 = 131.6$ by (23.9). B ignores ρ; A uses $1 - \rho$. The factor $1 - \rho^2$ is the product of the correlation matrix's eigenvalues 1.9 and 0.1: the small one drives the inverse, and shrinkage lifts it.

6. **B** is correct: the diagonal of $\boldsymbol{\nu}\mathbf{1}^\top + \mathbf{1}\boldsymbol{\nu}^\top$ is $2\nu_i$. A takes one of the two terms. C forgets the diagonal; each covariance with asset 3 falls by $\nu_3 + \nu_j$.

7. Section 4's proof gives (30.2). Here $\operatorname{tr}\Sigma/N = 2.5$, so $\|\Sigma - 2.5I\|_F^2 = 1.5^2 + 1.5^2 = 4.5$. By (18.4) the noise is $(16 + 1 + 25)/100 = 0.42$, and $c^* = 4.5/4.92 = 0.9146$.

8. Section 5 gives (30.4). Here $\hat{\Sigma}$ has entries 1, 1.6 and 4, and the unconstrained weights are $(4 - 1.6)/(1 + 4 - 3.2) = 133.3\%$ and −33.3%. At $\mathbf{w} = (1, 0)$ the KKT conditions hold with $\nu_0 = (\hat{\Sigma}\mathbf{w})_1 = 1$ and $\nu_2 = 1.6 - 1 = 0.6 \geq 0$, so the long-only weights are (1, 0). Then $\hat{\Sigma}_\nu$ has entries 1, $1.6 - 0.6 = 1$ and $4 - 1.2 = 2.8$, with minimum-variance weights $(2.8 - 1)/(1 + 2.8 - 2) = 100\%$ and 0.

9. Section 6 gives (30.5). Here $\mathbf{e}_{\text{eq}} = 4 \times (0.04 \times 0.5,\ 0.01 \times 0.5) = (8\%, 2\%)$, so $K\mathbf{e}_{\text{eq}} = -6\%$ and $K\hat{\Sigma}K^\top = 0.05$, and the factor $c(b - K\mathbf{e}_{\text{eq}})/K\hat{\Sigma}K^\top$ is $0.5 \times 0.06/0.05 = 0.6$. Then

$$\begin{aligned}
\mathbf{e}_{\text{BL}} &= (8\%, 2\%) + 0.6\,(-0.04,\ 0.01) \\
&= (5.6\%,\ 2.6\%), \\
\mathbf{f}_{\text{BL}} &= (50\%, 50\%) + \tfrac{0.6}{4}(-1,\ 1) \\
&= (35\%,\ 65\%).
\end{aligned}$$

10. The lab gives $\hat{c} = 0.9675$ both ways, the oracle 0.9851 and the ratio 2.14 (Example 4). The shares are 36% (0050), 21% (00646), 4% (00679B), 0.2% (00635U) and 39% (00642U). They show 0050's fat tails and 00642U's large variance.

11. The ratios are 0.423, 0.425 and 0.87686 (SEs 0.003, 0.003 and 0.00002); 1/N wins in 31.4% of samples (SE 0.5 points); 2,115 samples have $\mathbf{1}^\top\hat{\mathbf{f}} \leq 0$. In the model the best Sharpe ratio is well above 1/N's (0.880 against 0.327) and the assets are few, but the window is short. In the data the ex ante gap is unknown, since 0.880 is the window's in-sample optimum (Section 2's pitfall).

12. (a) At 30 returns ($T/N \approx 1.1$) the sample portfolio realizes 43.12% and the shrunk one 12.46%: shrink, or constrain (0.38 points better in experiment 3, SE 0.03). At 2,191 ($T/N \approx 78$) both realize 11.78%, and the constraint costs (12.14%). The fitted $\hat{c}$, κ and predictions are in-sample, the realized volatilities out-of-sample. (b) The check of (30.4) finds 19 binding constraints and weights that agree to 10^{-15}. (c) At 60 returns ($T/N \approx 2.1$) shrink or constrain: in experiment 3 both cut the mean true volatility from 15.51% to under 13.5%. Shrinkage pulls the variances together, and the constraint adds bias when the truth needs shorts. At 504 returns ($T/N \approx 101$) the covariances need little repair (R5 and R3 differ by 0.01 points). The means need one, a reference portfolio, which carries the prior.

GLOSSARY

Term	中文	Meaning
Plug-in portfolio	代入投資組合	Weights computed from estimates as if they were the truth, such as $\hat{\Sigma}^{-1}\hat{\mathbf{e}}/\delta$.
Estimation error	估計誤差	The difference between an estimate and the quantity it estimates.
Sampling error	抽樣誤差	The part of the estimation error that comes from a finite sample of a fixed model.
Shrinkage estimator	收縮估計量	A weighted average of an estimator and a target, such as $\hat{\Sigma}_c$ of (30.1).
Oracle weight	—	c^* of (30.2), the best nonrandom weight on the data, which needs the truth.
Ledoit–Wolf estimator	—	The shrunk matrix with $\hat{c}$ of (30.3).
Frobenius norm	Frobenius 範數	$\|M\|_F$, the square root of the sum of squared entries (LM18).
Condition number	條件數	$\kappa = \lambda_1/\lambda_N$ of a positive definite matrix; κ_c after shrinkage.
Adjusted covariance matrix	—	$\hat{\Sigma}_\nu$ of (30.4), whose unconstrained minimum-variance portfolio is the long-only one.
Reference portfolio	參考投資組合	The weights $\mathbf{w}_{\text{ref}}$ that Black–Litterman's prior starts from.
Implied excess returns	隱含超額報酬	$\mathbf{e}_{\text{eq}} = \delta\hat{\Sigma}\mathbf{w}_{\text{ref}}$, the means under which the reference is optimal.
View	觀點	A linear statement $K\mathbf{e} = b + \varepsilon$ about excess returns, with an error.
Resampled efficiency	—	Averaging optimal weights over draws of the inputs (Michaud and Michaud 2008).
Single index model	單一指標模型	The covariance matrix $\hat{\Sigma}_{\text{F}}$ of market betas and uncorrelated residuals.
Nearest correlation matrix	—	The PSD matrix with unit diagonal closest to a given one in the Frobenius norm.

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